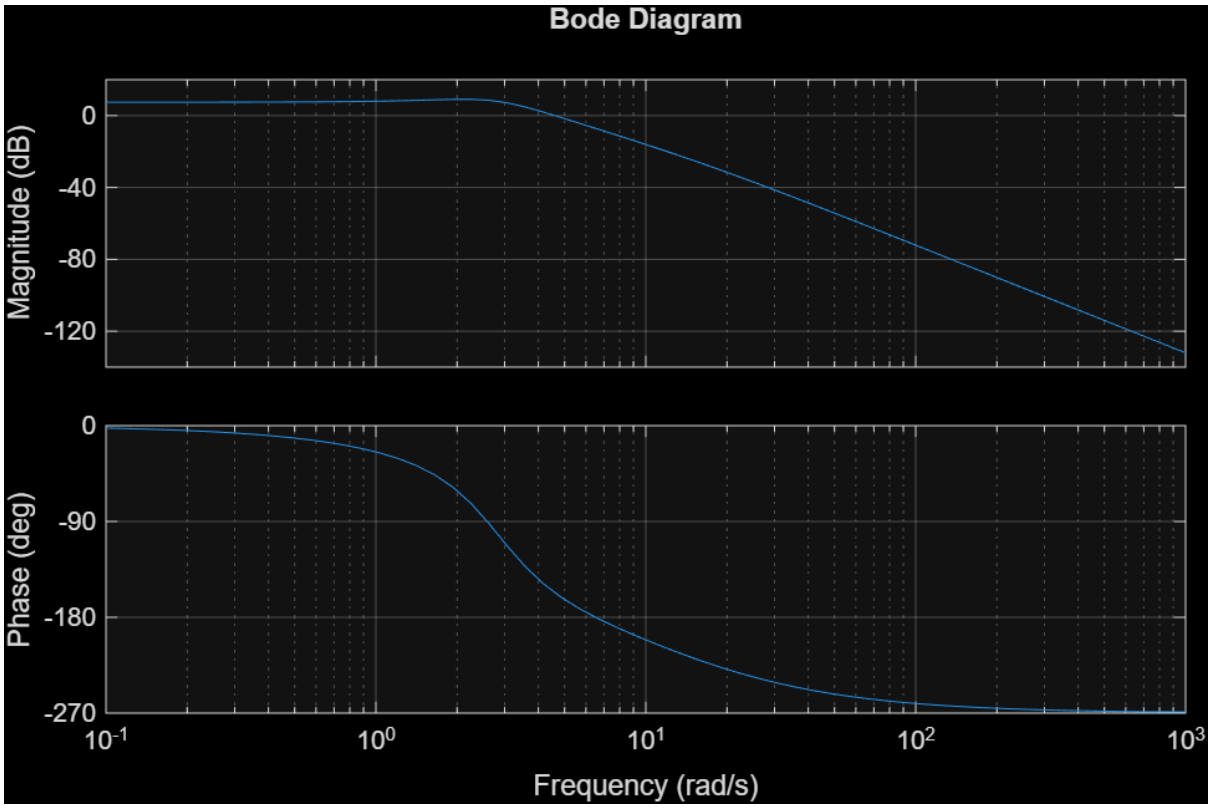
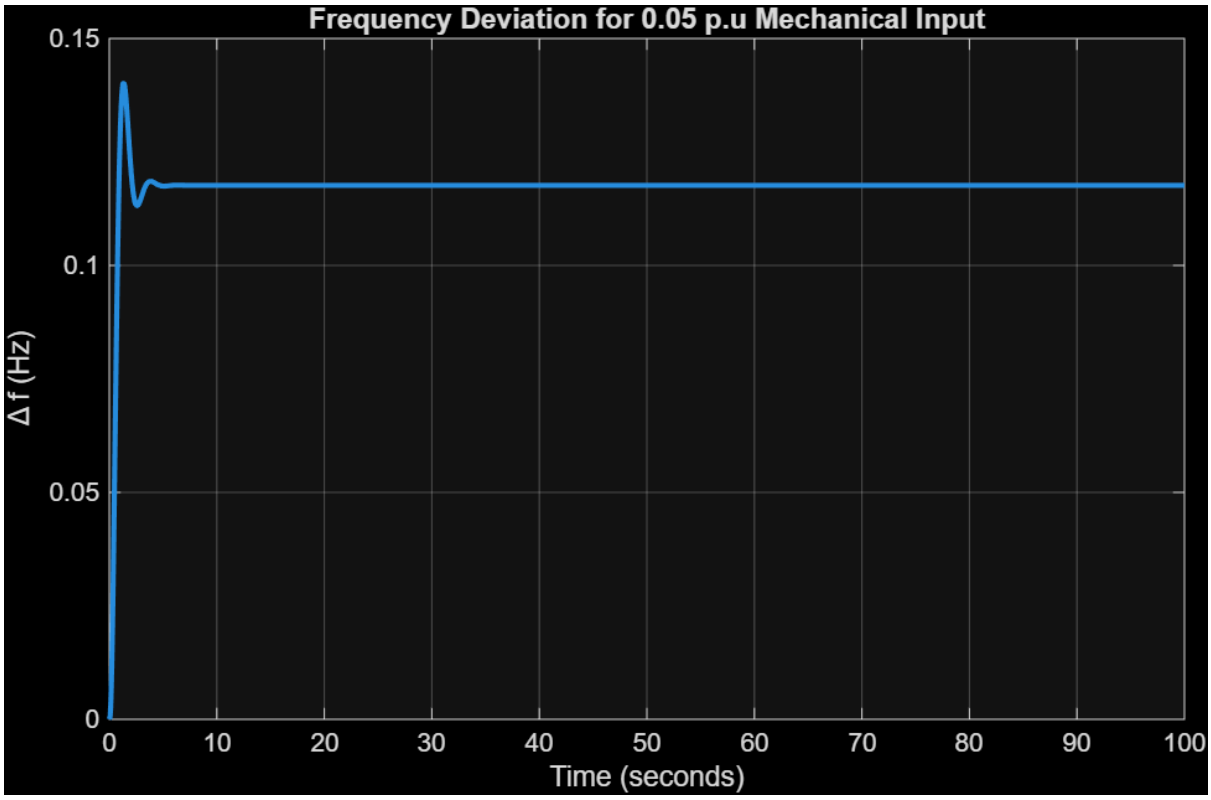
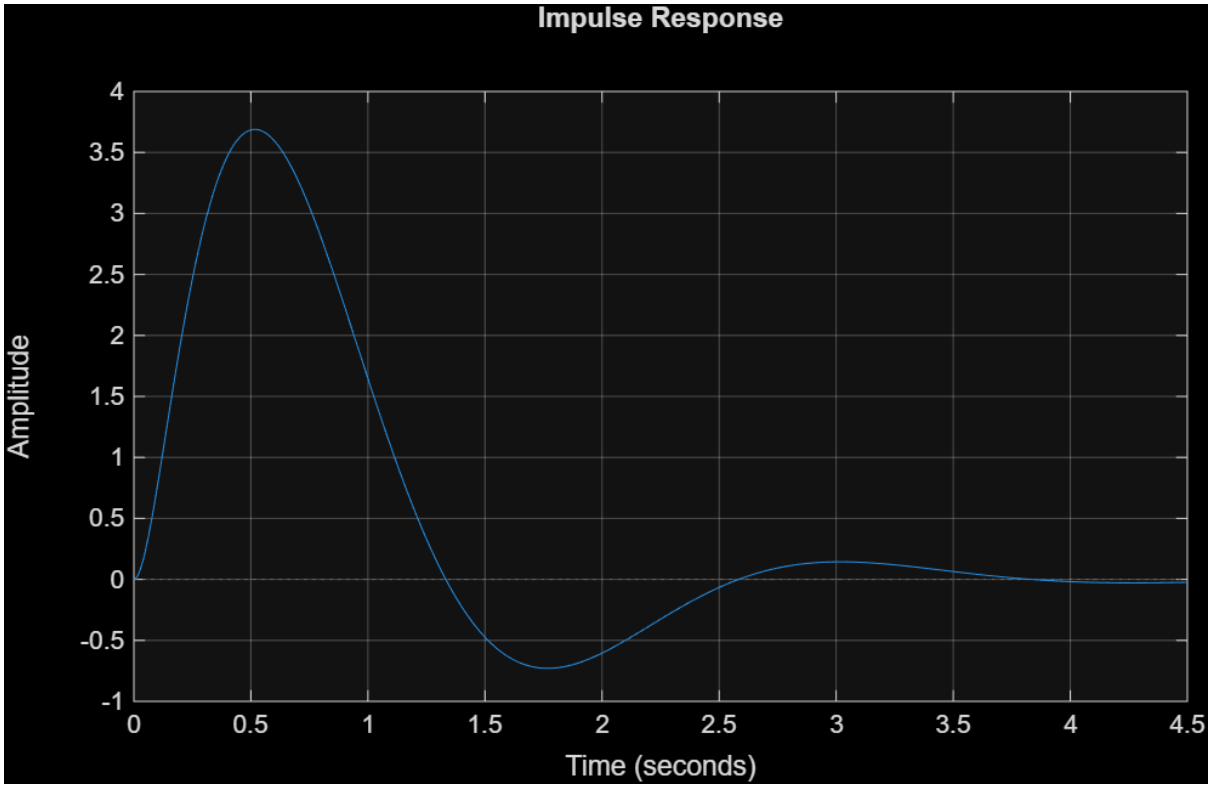
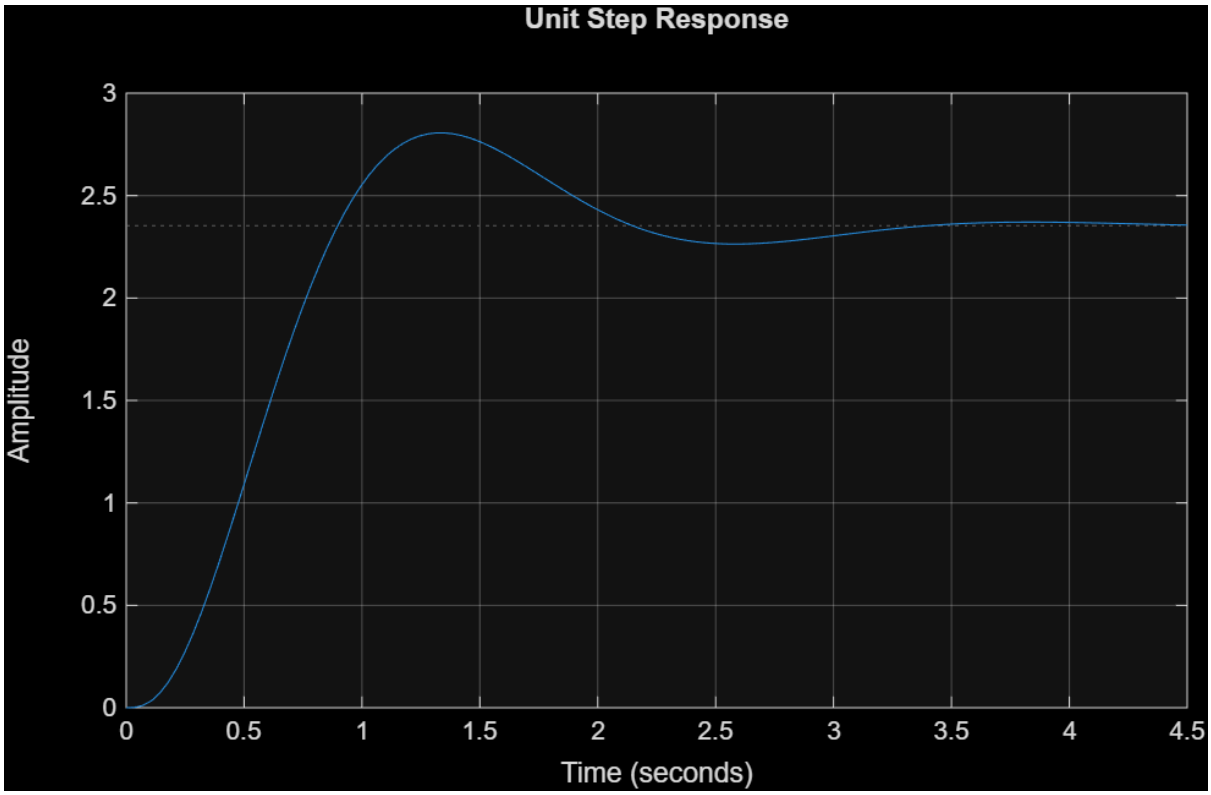
```
%% Single Area Load Frequency Control
clc;
clear;
close all;
%% System Parameters
Tg = 0.08;          % Governor Time Constant (s)
Tt = 0.30;          % Turbine Time Constant (s)
Tp = 20;            % Power System Time Constant (s)
Kp = 120;           % Power System Gain
R  = 2.4;           % Speed Regulation
%% Transfer Functions
s = tf('s');
Governor = 1/(Tg*s + 1);
Turbine = 1/(Tt*s + 1);
PowerSystem = Kp/(Tp*s + 1);
%% Open Loop
G = Governor*Turbine*PowerSystem;
%% Closed Loop with Governor Feedback
H = 1/R;
ClosedLoop = feedback(G,H);
disp('Closed Loop Transfer Function')
ClosedLoop;
%% Simulation Time
t = 0:0.01:100;
%% Mechanical Input Change
DeltaPm = 0.05;
u = DeltaPm*ones(size(t));
%% Controlled Response
[y,t] = lsim(ClosedLoop,u,t);
%% Plot
figure
```

```
plot(t,y,'LineWidth',2)
grid on
xlabel('Time (seconds)')
ylabel('\Delta f (Hz)')
title('Frequency Deviation for 0.05 p.u Mechanical Input')
%% Step Information
disp('Step Characteristics')
stepinfo(y,t)
%% Impulse Response
Figure
impulse(ClosedLoop)
grid on
title('Impulse Response')
%% Step Response
figure
step(ClosedLoop)
grid on
title('Unit Step Response')
%% Pole Zero Map
figure
pzmap(ClosedLoop)
grid on
%% Root Locus
figure
rlocus(ClosedLoop)
grid on
%% Bode Plot
figure
bode(ClosedLoop)
grid on
```


Bode Diagram
Magnitude (dB)
0
-40
-80
-120
Phase (deg)
0
-90
-180
-270
10-1
100
101
102
103
Frequency (rad/s)


Frequency Deviation for 0.05 p.u Mechanical Input
Δ f (Hz)
0.15
0.1
0.05
0
0
10
20
30
40
50
60
70
80
90
100
Time (seconds)


Impulse Response
Amplitude
Time (seconds)
4
3.5
3
2.5
2
1.5
1
0.5
0
-0.5
-1
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5


Unit Step Response
Amplitude
Time (seconds)
3
2.5
2
1.5
1
0.5
0
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5

Closed Loop Transfer Function

ClosedLoop =

$$\frac{120}{0.48\,s^3 + 7.624\,s^2 + 20.38\,s + 51}$$

Continuous-time transfer function.
Model Properties

Step Characteristics

ans = **struct** with fields:

RiseTime: 0.5703
TransientTime: 3.0161
SettlingTime: 3.0161
SettlingMin: 0.1068
SettlingMax: 0.1403
Overshoot: 19.2716
Undershoot: 0
Peak: 0.1403
PeakTime: 1.3300